

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1

PAPER 1 November 2018

2 hours 30 minutes

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2018 4004/1.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.

INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [].

1. Express 30 as a percentage of 60. [2]

2. Work out $4/3 \div 2/3$. [3]

3. Write 70000 in standard form. [2]

4. Overnight in Bulawayo the temperature was 2°C . By 14:00 it was 24°C . Find the rise. [2]

5. $n(A) = 13$, $n(B) = 11$, $n(A \cap B) = 3$. Find $n(A \cup B)$. [3]

6. Simplify $(3^2)^2$. Leave as a power of 3. [3]

7. Share \$22 in the ratio 4 : 7. [4]

8. Expand $(x + 4)(x + 2)$. [4]

9. Express 96 as a product of its prime factors. [4]

- 10.** Find the HCF and LCM of 12 and 42. **[4]**
- 11.** Make r the subject of $C = 2\pi r$. **[4]**
- 12.** Find the mean of 6, 12, 9, 12, 11. **[3]**
- 13.** Solve $3x + 3 < 12$. **[3]**
- 14.** Find the midpoint of (0, 8) and (3, 9). **[3]**
- 15.** Solve
$$\begin{cases} x + y = 6 \\ x - y = 0 \end{cases}$$
 [4]
- 16.** The n th term is $8 + 5(n - 1)$. Find the 8th term. **[3]**
- 17.** A triangular plot has base 10 m and perpendicular height 5 m. Find its area. **[3]**
- 18.** Factorise $x^2 - 16$. **[4]**
- 19.** $\tan 45^\circ = 1$. In a right-angled triangle the adjacent side to 45° is 8 cm. Find the opposite side. **[4]**
- 20.** A cuboid water tank measures 12 m by 6 m by 5 m. Find its volume. **[3]**
- 21.** Solve $6x + 7 = 25$. **[3]**

- 22.** A ladder of length 15 m leans against a wall. The foot is 9 m from the wall. How far up the wall does it reach? **[4]**
- 23.** A shirt in Bulawayo is marked \$120. In a sale it is 25% off. Sale price? **[4]**
- 24.** Write the equation of the line with gradient 3 passing through (0, 4). **[3]**
- 25.** Factorise $x^2 + 8x + 12$. **[4]**
- 26.** In triangle ABC, $\hat{A} = 55^\circ$ and $\hat{B} = 69^\circ$. Find $\hat{C}$. **[3]**
- 27.** $u = (3 ; 0)$, $v = (-2 ; 5)$. Find $2u + v$. **[4]**
- 28.** A regular polygon has 6 sides. Find each exterior angle. **[3]**
- 29.** A bag contains 5 red, 5 blue and 5 green beads. $P(\text{not green})$? **[3]**
- 30.** Circumference of a circle radius 21 cm. Take $\pi = 22/7$. **[4]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).